

Test Report

JSONPlaceholder – API Testing

API Under Test: <https://jsonplaceholder.typicode.com> | Tester: QA Portfolio Project | Date: September 11, 2026

1. Objective

This report documents API testing of the JSONPlaceholder REST API using Postman, covering the core HTTP methods (GET, POST, PUT, PATCH, DELETE), status code validation, response body verification, nested resource retrieval, and error handling for invalid requests.

2. Test Environment

API Base URL: <https://jsonplaceholder.typicode.com>
Tool: Postman
Test Type: Manual API testing (functional, status code, and response body verification)

3. Test Case Summary

ID	Method	Endpoint	Test Case	Expected	Actual	Status
TC01	GET	/posts	Get all posts	200, list of 100 posts	200, list of 100 posts	Passed
TC02	GET	/posts/1	Get a single existing post	200, post with id 1	200, correct post returned	Passed
TC03	GET	/posts/9999	Get a non-existent post	404, empty body	404, empty body {}	Passed
TC04	POST	/posts	Create a new post	201, echoes sent data + new id	201, returned title/body/userId + id 101	Passed
TC05	PUT	/posts/1	Full update of a post	200, full object replaced	200, all fields replaced as sent	Passed
TC06	PATCH	/posts/1	Partial update of a post	200, only sent field changes	200, only title changed, rest unchanged	Passed
TC07	DELETE	/posts/1	Delete a post	200, empty object	200, empty object {}	Passed
TC08	GET	/posts/1/comments	Get comments for a post	200, list of comments, postId = 1	200, list of comments, all postId = 1	Passed
TC09	GET	/posts/abc	Get post with invalid ID	404, empty body	404, empty body {}	Passed
TC10	GET	/users/1/todos	Get nested resource (todos)	200, list of todos for user 1	200, list of todos returned	Passed

4. Overall Results

Metric	Result
Total Test Cases Executed	10
Passed	10
Failed	0
HTTP Methods Covered	GET, POST, PUT, PATCH, DELETE
Pass Rate	100%

All 10 test cases passed across five HTTP methods. During execution, TC04, TC05, and TC06 initially returned incomplete response bodies because the request method and JSON body were not set correctly in Postman. This was identified through careful inspection of the response content (not just the status code), corrected, and re-verified - see note in Section 5.

5. Verification Note — PUT vs PATCH Behavior

TC05 (PUT) and TC06 (PATCH) both target the same resource (/posts/1) and return status 200, but demonstrate different underlying behavior. PUT performs a full replacement of the resource: the response reflects exactly the fields sent in the request. PATCH performs a partial update: only the field(s) included in the request body change, while all other fields retain their original values. This distinction was verified directly from the response bodies rather than assumed from the status code alone, which is a similar principle to the price-verification finding in Test Report #2 - correct behavior should always be confirmed against the actual data returned, not just a successful status code.

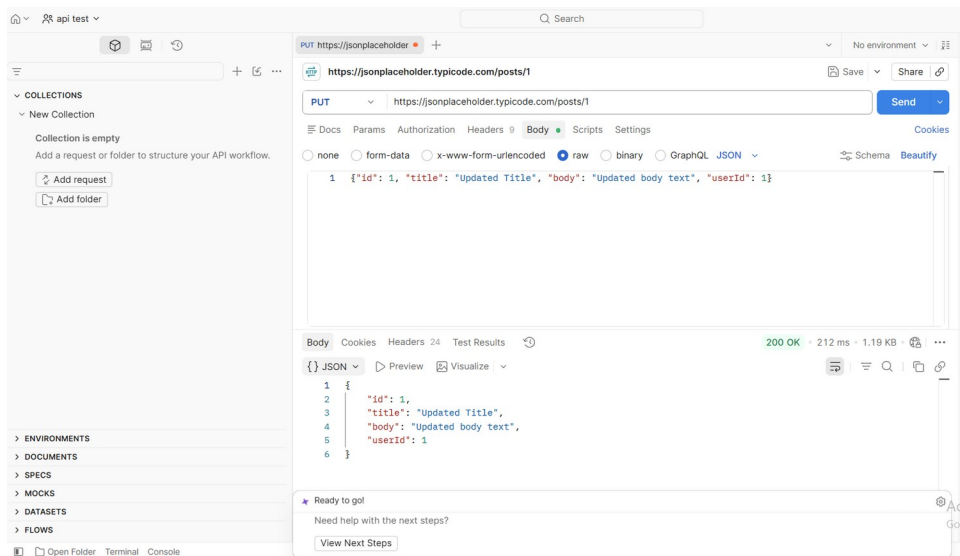


Figure 1 — TC05 (PUT): full object replaced in the response

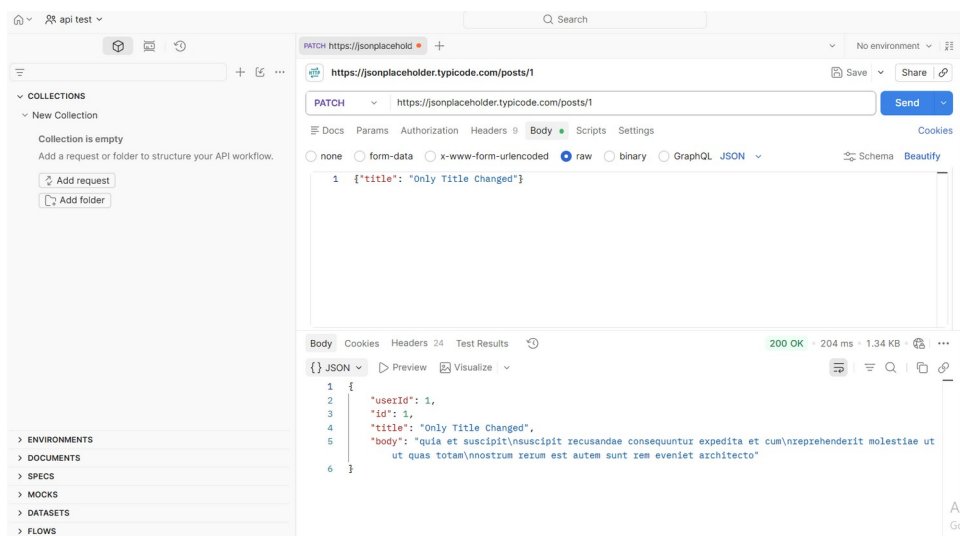


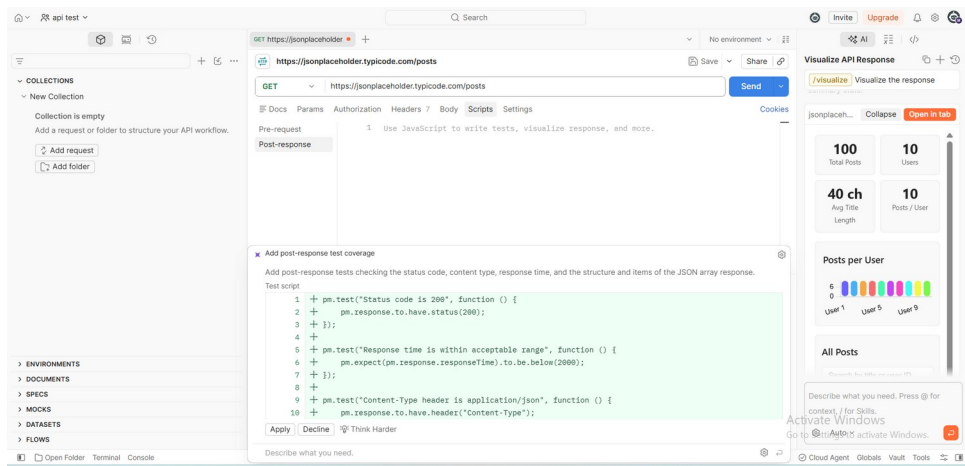
Figure 2 — TC06 (PATCH): only the title field changed, other fields preserved

6. Conclusion

The JSONPlaceholder API behaved as expected across all tested methods and endpoints, including correct error handling (404) for non-existent and invalid resource IDs. This project also reinforced a key testing principle applied throughout this portfolio: verifying actual response content, not just status codes, to catch both test execution errors and confirm correct application behavior.

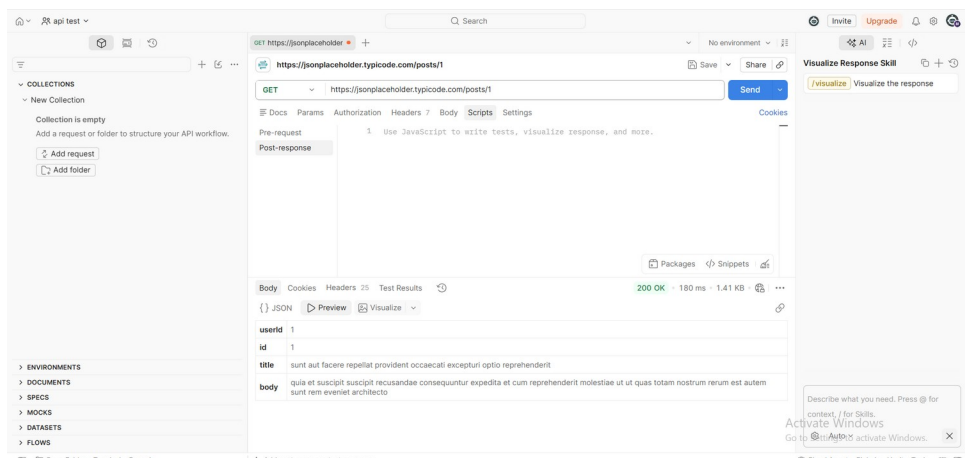
Appendix — Evidence Screenshots

TC01 — GET /posts (list of 100 posts)



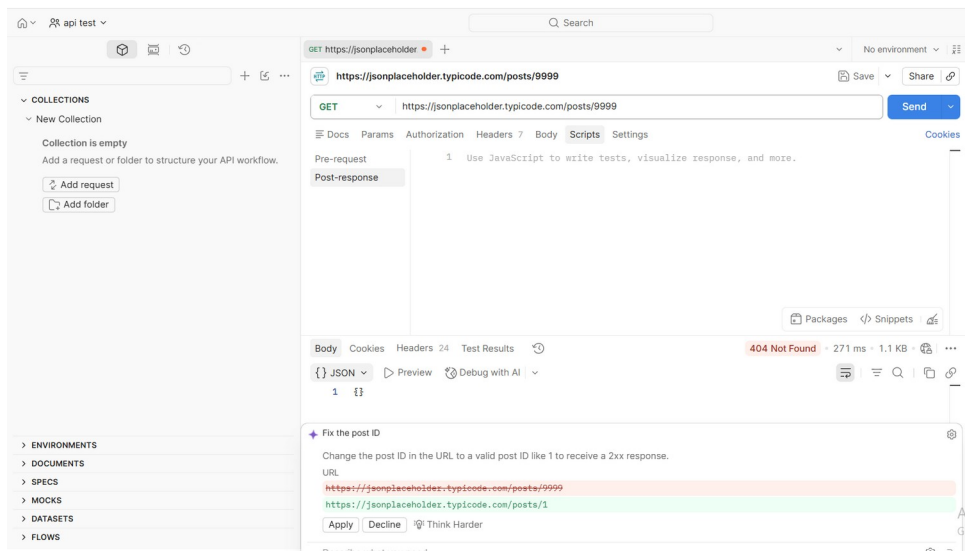
TC01 evidence

TC02 — GET /posts/1



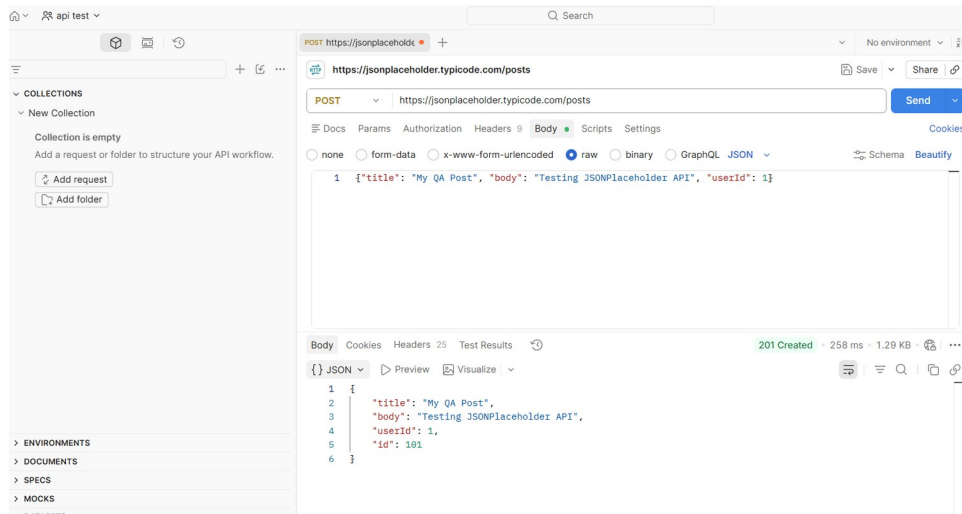
TC02 evidence

TC03 — GET /posts/9999 (404 Not Found)



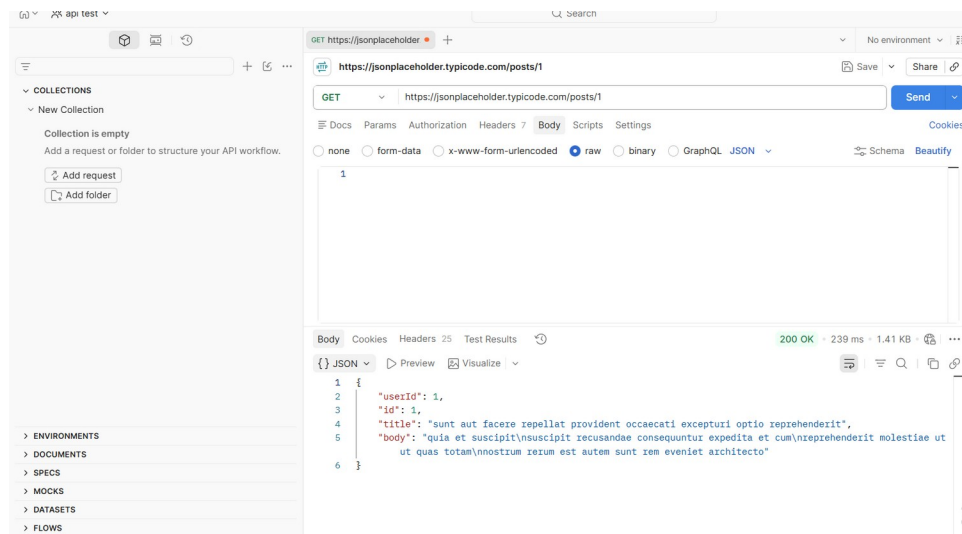
TC03 evidence

TC04 — POST /posts (create, corrected)



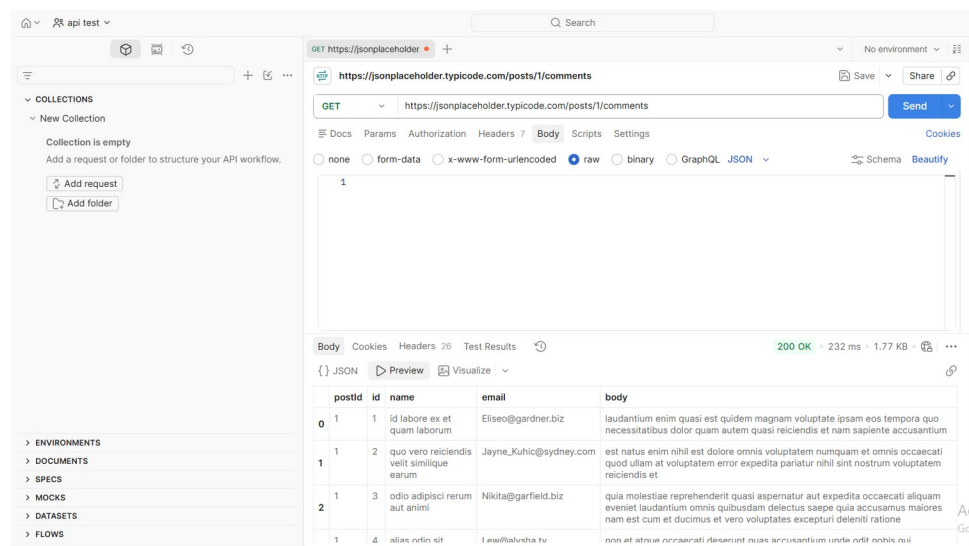
TC04 evidence

TC07 — DELETE /posts/1



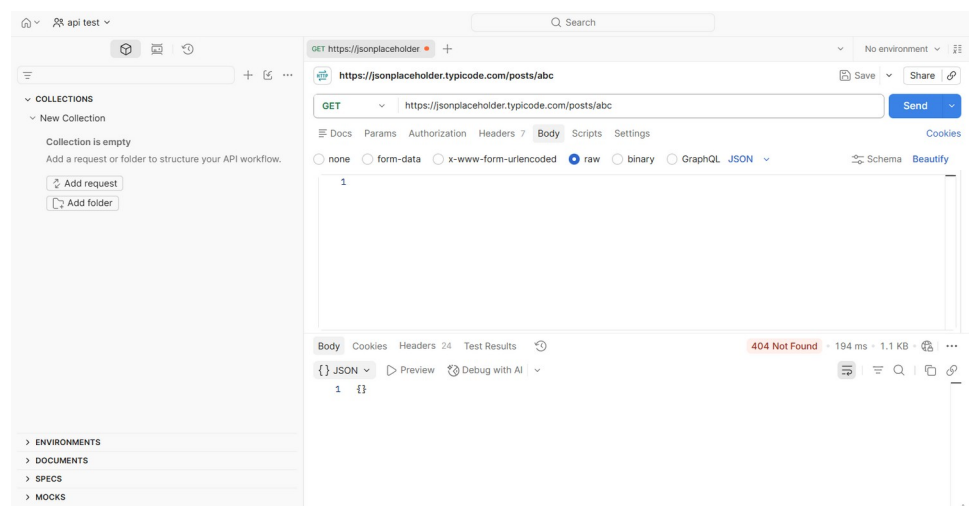
TC07 evidence

TC08 — GET /posts/1/comments



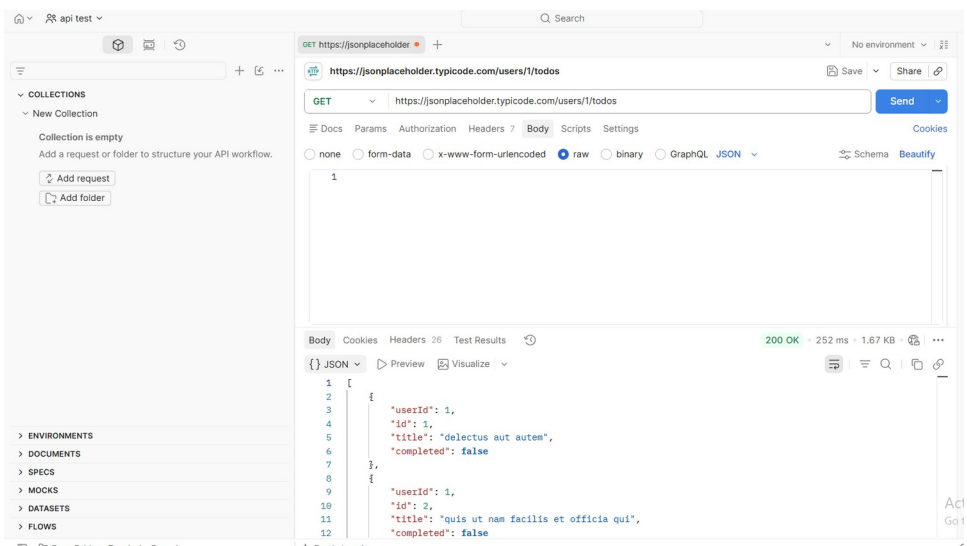
TC08 evidence

TC09 — GET /posts/abc (invalid ID, 404 Not Found)



TC09 evidence

TC10 — GET /users/1/todos (nested resource)



TC10 evidence